`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

We present `bispectrum`, an open-source PyTorch library that computes selective G-bispectral invariants for seven group/domain pairs as drop-in differentiable `nn.Modules`. The selective bispectrum reduces invariant cost from $O(|G|^2)$ to $O(|G|)$ while remaining complete for generic signals. Building the library required two theoretical contributions: (i) an augmented selective $\mathrm{SO}(3)$ invariant on S^2 achieving $\Theta(L^2)$ coefficients—the information-theoretic lower bound—instead of $O(L^3)$ for the full bispectrum (completeness proved for $4 \leq L \leq 100$), and (ii) diagnosing a previously undocumented inversion failure for the octahedral group. Across three benchmarks (C_8, octahedral, $\mathrm{SO}(3)$), the library delivers numerically exact rotation invariance ($\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage of complete bispectral invariants over incomplete alternatives in the low-data, moderate-capacity regime. The selective forward pass runs in sub-millisecond time on GPU for finite groups.

1 Introduction

Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce symmetry by construction, but they still require an invariant representation at the point where a prediction must no longer depend on orientation. In practice, invariant pooling remains a weak point: standard choices such as norm pooling, gate-based pooling, or max/sum aggregation are usable [Weiler and Cesa, 2019], but they do not preserve all equivariant information. The bispectrum offers a mathematically principled alternative, since for many groups it yields exact invariants with reconstruction guarantees [Kakarala, 1992].

The obstacle is practical rather than conceptual. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the finite-group and compact-group cases relevant to modern geometric deep learning. Even for groups where selectivity reduces the coefficient count from $O(|G|^2)$ to $O(|G|)$ [Mataigne et al., 2024], a user still needs a numerically stable implementation, a consistent API, inversion routines, tests, and benchmarked runtime behavior.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes two technical contributions that arose during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 with $O(L^2)$ coefficients (Section 3.4), and an analysis of the bootstrap inversion failure [Mataigne et al., 2024] for the octahedral group together with a partial workaround (Section 3.5).

Contributions.

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Table 1: Invariant nonlinearities for equivariant CNNs. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Sel. bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	Genericity assumption

1. **Software contribution.** A PyTorch library implementing bispectral invariants for seven group/domain pairs with a consistent `nn.Module`-style interface, GPU-optimised forward passes (<1 ms for finite groups), precomputed Clebsch–Gordan buffers, and parallel construction routines (Section 3.6).
2. **Augmented selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct and implement an augmented selective bispectrum for $\mathrm{SO}(3)$ acting on S^2 that uses $\Theta(L^2)$ coefficients instead of the full $\Theta(L^3)$, matching the orbit-space lower bound up to a constant factor. The invariant combines scalar bispectral entries with Clebsch–Gordan power spectrum entries (a degree-4 complement needed for real signals). Completeness is proved for $4 \leq L \leq 100$ via a combination of exact rational certificates and a closed-form linear bootstrap family (Appendix A); the verification extends to any desired L by re-running a single script.
3. **Octahedral inversion analysis.** An analysis of why the standard selective-bispectrum bootstrap inversion fails for the octahedral group, together with a bootstrap + Levenberg–Marquardt approach that achieves local convergence but not reliable global recovery.
4. **Empirical validation.** A task-driven evaluation of the library as an invariant pooling primitive across three group/domain settings, showing that complete bispectral invariants provide a data-efficiency advantage in the low-data, moderate-capacity regime while converging to incomplete alternatives at high capacity.

2 Mathematical Background

2.1 Equivariant features and invariant maps

A neural network layer is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Stacking equivariant layers preserves this property, but any task that requires a prediction independent of orientation—classification, regression, detection—must eventually map equivariant features to *invariant* ones.

Table 1 summarises the trade-offs. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with irreducible representations $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i, and let $\hat{f}(\rho_i) = \sum_{g \in G} f(g)\, \rho_i(g)$ denote the group Fourier transform. Given the Clebsch–Gordan decomposition $\rho_i \otimes \rho_j \cong \bigoplus_k m_{ij}^k \rho_k$ with intertwining matrix C_{ij}, the *bispectrum* is

$$\beta_{\rho_i,\rho_j} = C_{ij}^\top \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right) C_{ij}\, \hat{f}(\rho_k)^\dagger. \tag{1}$$

Each β_{ρ_i,ρ_j} is invariant under $f \mapsto f \circ g$ and, taken over all pairs (i, j), the bispectrum is a *complete* invariant: it separates orbits for generic signals [Kakarala, 1992]. However, the full set has $O(|G|^2)$ coefficients.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a BFS traversal of the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients that remain complete. This reduction from quadratic to linear is the key to practical deployment: for example, the octahedral group drops from 576 to 172 coefficients.

3 The `bispectrum` Library

3.1 Design and implementation

Every group/domain pair is exposed as a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`). Modules accept raw real-valued inputs, handle the Fourier transform internally, and default to $O(|G|)$ selective coefficients (`selective=False` gives the full $O(|G|^2)$ set). CG matrices, DFT kernels, and Bessel roots are precomputed at construction and stored as non-learnable buffers, so `.to(device)` works transparently and forward passes incur zero recomputation. Where available, `bsp.invert(beta)` returns the reconstructed signal up to group-action indeterminacy. The only dependencies are PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel and cached to disk; all forward methods are batch-native, using `torch.einsum`/`torch.bmm` for non-abelian modules and `torch.fft` for abelian ones.

3.2 Supported modules

Table 2: Current scope of the `bispectrum` library.

Module	Group / Domain	Output mode
`CnonCn`	C_n on C_n	selective + full
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full
`TorusOnTorus`	product cyclic groups on torus	selective + full
`DnonDn`	D_n on D_n	selective
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full
`OctaonOcta`	octahedral group	selective

3.3 Example API

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

3.4 Augmented selective $\mathrm{SO}(3)$ bispectrum on S^2

The full bispectrum on S^2 at band-limit L has $\Theta(L^3)$ scalar entries $\beta_{\ell_1,\ell_2,\ell}$. For finite groups, selectivity reduces $O(|G|^2)$ to $O(|G|)$, but on S^2 the BFS chain $\beta_{1,0,1}, \beta_{1,1,2}, \ldots$ provides only $\sim 3L$ scalars—far too few for $(L{+}1)^2$ unknowns. Indeed, the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components. We achieve this lower bound.

Theorem 1 (Augmented selective $\mathrm{SO}(3)$ invariant on S^2)**.** *For $f : S^2 \to \mathbb{R}$ band-limited at degree $4 \le L \le 100$, there exists an augmented invariant $\Phi_{\mathrm{aug}}(f)$ consisting of $\Theta(L^2)$ scalars—a subset $\mathcal{S}_\beta$ of scalar bispectral entries $\beta_{\ell_1,\ell_2,\ell}$ together with a subset $\mathcal{S}_P$ of CG power spectrum entries $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$—that is a complete $\mathrm{SO}(3)$-invariant for Lebesgue-almost-every signal. The output size $\Theta(L^2)$ matches the dimension $(L{+}1)^2 - 3$ of the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ up to a constant factor.*

Remark 2. *The completeness proof (Appendix A) proceeds by gauge-fixing, seed recovery at degrees ≤ 3 (finite fibre), degree-4 fibre reduction via parity-breaking bispectral entries, and a linear bootstrap for $\ell \geq 5$. All rank and consistency conditions are verified at explicit rational or deterministic witnesses for each $\ell \leq 100$ and promoted to generic statements by algebraic-variety arguments. The structural form of the bootstrap is uniform for $\ell \geq 8$; extending the verification beyond $L = 100$ requires only re-running a single script.*

Why augmentation is necessary. For *complex* signals, the scalar bispectrum alone suffices: the linear bootstrap matrix has full complex rank $2\ell{+}1$ at each degree (verified at explicit witnesses up to $\ell = 100$). However, for *real* signals, the reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ creates algebraic dependencies in the bootstrap matrix that *cannot* be resolved by choosing different bispectral triples. At $\ell = 4$, for example, the bootstrap matrix has complex rank 4 instead of $9 = 2\ell{+}1$, and this holds even for the *full* bispectrum (all valid triples). The CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$ capture the complementary *magnitude* information that the phase-sensitive scalar bispectral entries miss, and together they achieve full Jacobian rank.

The selective index set. The bispectral component $\mathcal{S}_\beta$ selects up to $2\ell{+}1$ triples per degree from four categories: **(1) Chain** (ℓ_1, ℓ_2, ℓ) with $\ell_1 \leq \ell_2 < \ell$; **(2) Cross** (ℓ_1, ℓ, ℓ') with $\ell_1, \ell' < \ell$; **(3) Power** $(0, \ell, \ell)$; **(4) Self-coupling** (ℓ, ℓ, ℓ') with ℓ' even. Entries with $\ell_1 = \ell_2$ and odd ℓ vanish identically for real signals (CG antisymmetry of the symmetric square) and are excluded. All *even* self-coupling entries $\beta_{\ell,\ell,\ell'}$ with $2 \leq \ell' \leq \ell$ are included unconditionally; they are needed for full Jacobian rank in the seed system (Lemma 27). The invariant is further augmented with CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$, chosen to maximise the Jacobian rank of the combined invariant (Algorithm 1).

Table 3: Per-degree entry counts in the augmented selective $\mathrm{SO}(3)$ invariant. The "Bispec" column counts scalar bispectral entries including mandatory even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$); entries that vanish for real signals are excluded. The "CG Power" column counts CG power spectrum augmentations. At $\ell = 1$, $\beta_{1,1,0}$ is excluded (redundant with power). At $\ell = 2$, $\beta_{1,2,1}$ is excluded (proportional to $\beta_{1,1,2}$ after gauge-fixing).

ℓ	Bispec	CG Power	Total
0	1	0	1
1	1	0	1
2	3	1	4
3	7	4	11
4	12	5	17
$\ell \geq 5$	$\sim 2\ell{+}3$	$\sim \ell$	$\sim 3\ell$

Proof sketch. The proof establishes completeness of Φ_{aug} through gauge fixing and degree-by-degree recovery. *Base cases* ($\ell \leq 1$): $\mathbf{F}_0$ is recovered from $\beta_{0,0,0} = (a_0^0)^3$, and $\mathbf{F}_1$ from $\beta_{0,1,1}/a_0^0 = \|\mathbf{F}_1\|^2$ plus gauge-fixing (align $\mathbf{F}_1$ with the z-axis, fix the azimuthal phase of a_2^1). *Seed recovery* ($\ell = 2, 3$): the augmented seed system—scalar bispectral entries plus CG power entries—has full Jacobian rank 11 in 11 unknowns, yielding a finite fibre containing T_R-pairs, where $T_R\colon (\theta, \varphi) \mapsto (\theta, -\varphi)$ is azimuthal reflection. *Degree-4 fibre reduction*: the degree-4 augmented entries include parity-breaking bispectral entries ($\beta_{2,3,4}$ with odd $\ell_1{+}\ell_2{+}\ell$) whose imaginary parts flip sign under T_R. These eliminate all spurious seed solutions and resolve the T_R ambiguity, leaving a unique seed. *Bootstrap* ($\ell \geq 5$): the linear bootstrap has full complex rank at each degree, uniquely determining $\mathbf{F}_\ell$ from lower-degree coefficients. The computational certificates use double-precision arithmetic with well-separated thresholds (residual gaps of $\geq 10^{13}$ relative to machine epsilon); they can be upgraded to interval arithmetic if desired. The full proof is in Appendix A.

3.5 Octahedral inversion as an implementation case study

The bootstrap inversion of Mataigne et al. [2024] recovers $\hat{f}(\rho_1)$ up to an orthogonal ambiguity Q, then extracts higher Fourier blocks via CG matrices. For C_n and D_n the ambiguity is low-dimensional and absorbed by the CG structure, but for the octahedral group O the ambiguity is

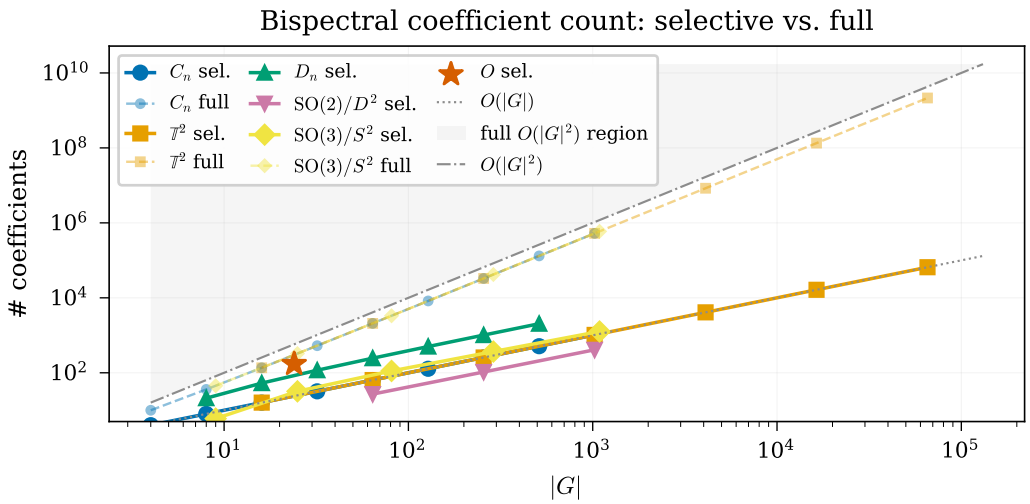

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension.

$Q \in \mathrm{SO}(3)$ and the CG matrices develop $O(1)$ cross-block leakage for generic $Q \notin O$, corrupting higher blocks. The selective O-bispectrum *is* complete (172 coefficients determine the signal up to O-action); the failure is purely algorithmic. We partially address it by combining bootstrap initialisation with Levenberg–Marquardt refinement of $\min_f \|\beta^{\mathrm{sel}}(f) - \beta^{\mathrm{sel}}_{\mathrm{target}}\|^2$, using multi-start randomisation over Q. The LM residual converges in <50 iterations, but the overall signal recovery rate remains modest (5.7% with 4 restarts), reflecting the difficult landscape created by CG leakage. The full analysis is in Appendix B.

3.6 Computational benchmarks

Table 4 and Figure 1 profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for forward, 4 for inversion, median wall-clock via `torch.utils.benchmark`).

Table 4: Library benchmark summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "–" = not implemented. For continuous groups, $|G| = \infty$; coefficient counts refer to the band-limited representation at the tested resolution.

Module	G	$\lvert G\rvert$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.15	9.17	8.56
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	524800	0.08	0.71	68.02
`DnonDn`	D_{32}	64	245	–	2.52	–	42.33
`SO2onDisk`	SO(2)	∞	411	–	1.94	–	17.69
`SO3onS2`	SO(3)	∞	430	–	26.54	–	–
`OctaonOcta`	O	24	172	–	0.70	–	2172.76

The selective forward pass is sub-millisecond for all abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms, SO(2)/disk: 1.94 ms) and under 1 ms for the octahedral group (0.70 ms), making it negligible relative to backbone computation. SO(3) on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at $|G|$=1,024 uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (9–68 ms) and practical for the octahedral group (2.17 s via Levenberg–Marquardt). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix C).

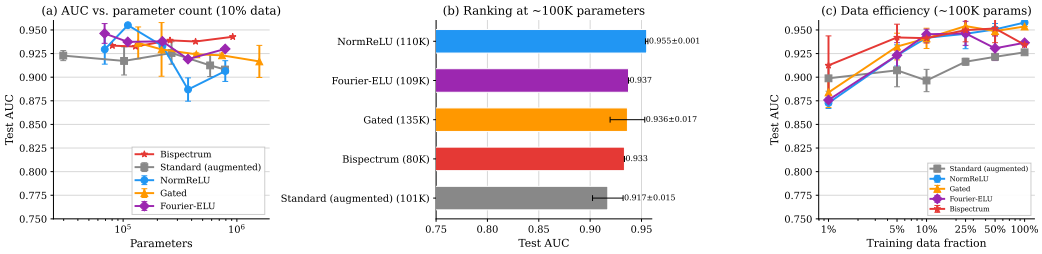

Figure 2: PCam Pareto analysis (single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics are in Table 5.

Table 5: PCam results at matched ∼100K parameters (10% training data, mean ± std over 3 seeds). All equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	$.896 \pm .010$	$.826 \pm .012$
Gated	136K	$.941 \pm .009$	$.870 \pm .004$
Bispectrum (`CnonCn`)	128K	$.941 \pm .004$	$.872 \pm .006$
Fourier-ELU	110K	$\mathbf{.945 \pm .004}$	$.855 \pm .006$
NormReLU	110K	$.942 \pm .004$	$\mathbf{.858 \pm .007}$

4 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 4.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 4.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 4.3).

4.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`$(n{=}8)$. An augmented standard CNN serves as a non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the data with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix D.

Results. Table 5 and Figure 2 present the results at matched ∼100K parameters.

At matched capacity (∼100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896 ± 0.010). Fourier-ELU leads slightly at 0.945 ± 0.004, followed by NormReLU (0.942), bispectrum and gated (both 0.941). Notably, the bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004), while gated has the highest (± 0.009), indicating that complete or near-complete invariants yield more reproducible training. At 1% training data, the bispectrum leads all methods (AUC 0.912 ± 0.025), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with the hypothesis that complete invariants provide a stronger inductive bias when data is scarce. Our best equivariant model (Fourier-ELU, 110K, AUC 0.945 on 10% data) approaches the published P4M-DenseNet of Veeling et al. [2018] (119K, AUC 0.963 on 100% data), despite using 10× less training data.

Table 6: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005

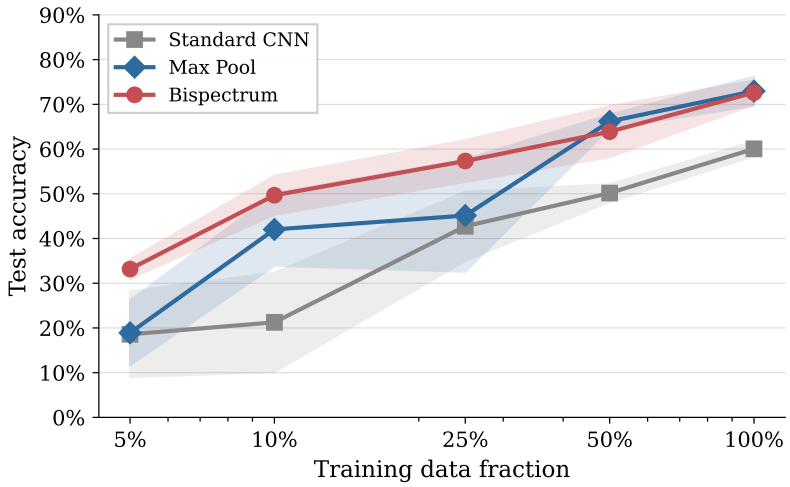

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands: ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime.

4.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models differ only in the terminal invariant pooling: a non-equivariant augmented CNN (16K params), norm pool (375K), max pool (374K), and the selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 6 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve numerically exact rotation invariance ($\sigma_{\text{rot}} \leq 0.0005$), while the augmented CNN drops $\sim$2.5pp under rotation ($\sigma_{\text{rot}} = 0.012$).

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%, with high seed variance $\pm 9.9\%$). At 463K parameters, the bispectrum also outperforms Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8% at 172K).

Data efficiency. When we reduce the training set (Figure 3), the bispectrum's advantage becomes clear: at 5% data (48 samples) it reaches $33.2 \pm 2.2\%$ vs. $18.9 \pm 7.2\%$ for max pool. Only at full data does max pool catch up (73.0% vs. 72.6%). The complete invariant map provides a stronger inductive bias that partially compensates for limited training data. However, at higher capacity (6M parameters, Appendix E), max pool overtakes the bispectrum (78.5% vs. 68.5%), suggesting the completeness advantage is specific to the data-limited regime.

Table 7: Spherical MNIST results (mean $\pm$ std over 3 seeds). Cohen et al. reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
Bispectrum + MLP (ours)	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

4.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST Cohen et al. [2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 384 complex invariant coefficients (768 real features after log-compression), which are fed to a 3-layer MLP (232K parameters total). The completeness proof (Appendix A) covers $4 \leq L \leq 100$, so $\ell_{\max} = 15$ is within the verified range. Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with a larger MLP (166K), isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 7 and Figure 4 report the results.

Numerically exact $\mathrm{SO}(3)$ *invariance.* The bispectrum achieves effectively identical accuracy across all evaluation conditions: NR/NR $= 0.950$, R/R $= 0.949$, NR/R $= 0.951$. The standard deviation under random rotations is $\sim 10^{-4}$—effectively zero. Training on rotated vs. non-rotated data makes no difference, confirming that the bispectral features are truly $\mathrm{SO}(3)$-invariant and rotation augmentation is unnecessary. In contrast, the standard CNN achieves 99.1% on NR/NR but collapses to 13.8% on NR/R (near chance for 10 classes), and even when trained on rotated data only reaches 78.1% with substantial variance ($\sigma_{\text{rot}} \approx 0.04$ under random rotations).

Completeness gap. The power spectrum is also rotation-invariant ($\sigma_{\text{rot}} = 0.0003$) but achieves only 78% accuracy—17.2 percentage points below the bispectrum. To rule out a capacity confound, we scale the power spectrum MLP to 166K parameters. This 15× capacity increase yields only +1.4pp (78.0% → 79.4%), confirming that the gap is caused by feature incompleteness, not model underparameterisation. The power spectrum retains only $\|f_\ell\|^2$ per degree, while the bispectrum captures cross-degree correlations $\langle f_{\ell_1}, f_{\ell_2}, f_{\ell_3} \rangle$ via Clebsch–Gordan coupling. For digit classification, these phase relationships carry substantial discriminative information that additional downstream capacity alone does not appear sufficient to recover.

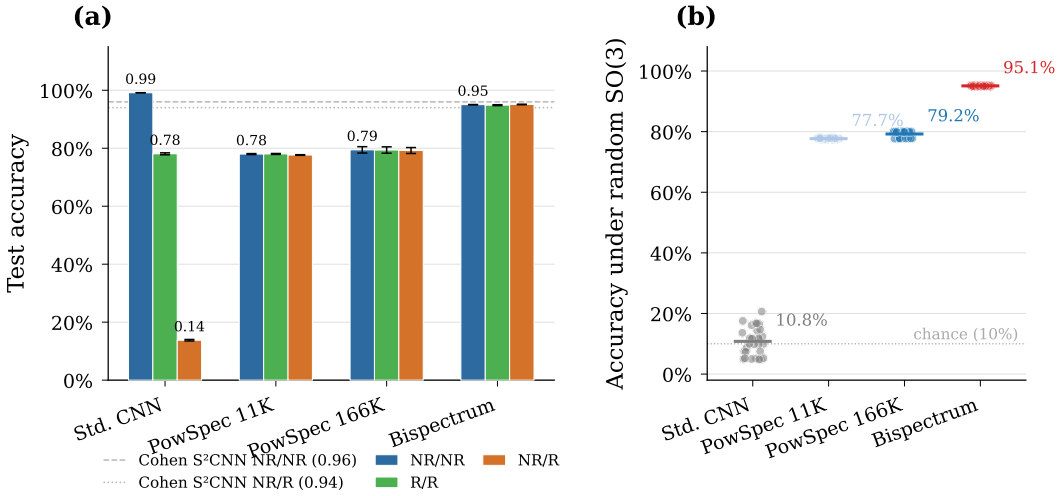

Figure 4: Spherical MNIST analysis. **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the bispectrum at 232K reaches 95.0%.

Comparison with Cohen et al. Cohen's spherical CNN Cohen et al. [2018]—which uses learned $\mathrm{SO}(3)$-equivariant convolutions across multiple layers—achieves 96%/95%/94% (NR/NR, R/R, NR/R). Our bispectrum reaches 94.8%/94.8%/94.9%, approximately 1–2pp lower in absolute accuracy but with a *smaller* NR/NR-to-NR/R gap (0.0pp vs. 2.0pp). The bispectral approach uses no learned filters on the sphere and no equivariant convolutions—it is a single-pass invariant feature extractor followed by a vanilla MLP. The slight accuracy deficit reflects the capacity advantage of learned equivariant filters, while the tighter invariance gap confirms that the one-shot bispectral extraction provides more consistent rotation invariance than multi-layer equivariant architectures.

5 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively. They implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with either framework.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalises to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction, and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives.

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2024] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The augmented selective bispectrum achieves completeness in a single $O(L^2)$ pass (for $\mathrm{SO}(3)$) or $O(|G|)$ pass (for finite groups), avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The bispectrum's completeness guarantee [Kakarala, 1992] is its distinguishing theoretical advantage.

6 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required two technical contributions during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $O(L^2)$ coefficient count (Theorem 1), and an analysis of the octahedral bootstrap inversion failure with a partial Levenberg–Marquardt workaround (Section 3.5). Table 4 confirms that the selective forward pass adds negligible overhead to training pipelines ($<$1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the complete invariant map provides a stronger inductive bias that partially compensates for limited training data.

When not to use the bispectrum. The data-efficiency advantage of complete invariants is most pronounced in the low-data, moderate-capacity regime. At high model capacity (Section 4), incomplete alternatives such as max pooling can match or surpass bispectral pooling, likely because the network learns to compensate for missing invariant information. Completeness is a property of the *map*, not of downstream task performance: a complete invariant provides a stronger inductive bias, but does not guarantee superior accuracy on every task. For $\mathrm{SO}(3)$ on S^2, computational cost grows as $O(L^2)$ per forward pass; at very large band-limits this may become a bottleneck relative to simpler pooling strategies.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness proof relies on a genericity assumption (the signal must avoid a measure-zero algebraic variety) and the bootstrap rank verification covers $L \leq 100$; a structural all-L proof and a full characterisation of the degenerate set remain open. The octahedral inversion achieves only a 5.7% signal recovery rate. Our experimental comparisons do not include tensor-product architectures (e3nn, MACE) or wavelet scattering, which target different input assumptions and would require non-trivial adaptation to the benchmarks used here.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; a structural proof of bootstrap rank for all L; numerical conditioning analysis at large band-limits; integration with `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022]; and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix provides the complete proof of Theorem 1 from the main text, establishing completeness of the augmented selective $\mathrm{SO}(3)$-bispectrum on S^2 for band-limits $4 \leq L \leq 100$ (extendable to any desired L by re-running the verification script).

A.1 Setup and notation

Definition 3 (Band-limited signal on S^2)**.** *A signal $f : S^2 \to \mathbb{R}$ is* band-limited *at degree L if its spherical harmonic expansion*

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{2}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 4 (Rotation action)**.** *Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\, \mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.*

Definition 5 (Bispectrum on S^2)**.** *For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$, the* bispectral coefficient *is*

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \, \overline{a_\ell^m}, \tag{3}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m \rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar (real for real f).

Definition 6 (CG power spectrum)**.** *For a triple* (ℓ_1, ℓ_2, ℓ) *satisfying the triangle inequality, the* CG power spectrum entry *is*

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Big| \sum_{\substack{m_1,m_2 \\ m_1+m_2=m}} C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2} \Big|^2. \tag{4}$$

This is a degree-4 *polynomial in the signal coefficients (real and non-negative for real* f*) and* $\mathrm{SO}(3)$*-invariant: the CG-coupled product transforms as* $D^\ell(g)$*, and* $\|\cdot\|^2$ *is invariant under unitary transformations.*

Proposition 7 (Size of the full bispectrum)**.** *The number of admissible triples* (ℓ_1, ℓ_2, ℓ) *with* $0 \le \ell_1, \ell_2, \ell \le L$ *satisfying the triangle inequality* $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$ *is* $\Theta(L^3)$.

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 8 (Invariance)**.** *Each* $\beta_{\ell_1,\ell_2,\ell}$ *is* $\mathrm{SO}(3)$*-invariant: the Wigner matrices satisfy* $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ *under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by* $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 9 (Dimensional lower bound)**.** *Any complete* $\mathrm{SO}(3)$*-invariant for band-limited signals on* S^2 *at band-limit* L *must have at least* $(L+1)^2 - 3$ *functionally independent components.*

Proof. The orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$. Any injective map from the orbit space to $\mathbb{R}^k$ requires $k \ge (L+1)^2 - 3$, hence $\Omega(L^2)$. □

Remark 10. *The chain* $\beta_{1,\ell,\ell+1}$ *for* $\ell = 0, \ldots, L-1$ *provides only* $\sim 3L$ *scalar constraints (since each chain entry couples* $\mathbf{F}_1 \in \mathbb{C}^3$ *with one known* $\mathbf{F}_\ell$*). This cannot recover* $(L+1)^2 \approx L^2$ *unknowns.*

A.3 The selective index set

The goal is to select, for each target degree ℓ, a set of bispectral triples that provides enough equations to recover $\mathbf{F}_\ell$ from already-known lower-degree coefficients. Since $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom, we need at most $2\ell + 1$ entries. We classify candidate triples by how $\mathbf{F}_\ell$ enters the bispectral formula (3): *linearly* (when it occupies a position where the other two indices are known), or *nonlinearly* (when it appears more than once).

Definition 11 (Selective bispectrum on S^2)**.** *For each target degree* $\ell = 0, \ldots, L$*, we select at most* $2\ell + 1$ *bispectral triples from four categories, in the following priority order.*

Category 1 (Chain). *Triples* (ℓ_1, ℓ_2, ℓ) *with* $\ell_1 \le \ell_2 < \ell$ *and* $\ell_1 + \ell_2 \ge \ell$*. Linear in* $\mathbf{F}_\ell$*: the unknown* $\overline{a^m_\ell}$ *appears once, multiplied by known lower-degree coefficients.*

Category 2 (Cross). *Triples* (ℓ_1, ℓ, ℓ') *with* $1 \le \ell_1 < \ell$ *and* $\ell - \ell_1 \le \ell' < \ell$*. Here* $\mathbf{F}_\ell$ *appears in the second (tensor-product) position, yielding equations linear in* $a^{m_2}_\ell$ *with known* $a^{m_1}_{\ell_1}$ *and* $a^m_{\ell'}$.

Category 3 (Power). *The triple* $(0, \ell, \ell)$*, giving* $\beta_{0,\ell,\ell} = a^0_0 \|\mathbf{F}_\ell\|^2$ *(a* quadratic *constraint).*

Category 4 (Self-coupling). *Triples* (ℓ, ℓ, ℓ') *with* $0 \le \ell' < \ell$*. Quadratic in* $\mathbf{F}_\ell$*. Selected only for the remaining deficit at low degrees.*

Algorithm 1 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    for $a = 1, 2, 3, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates {$\beta_{1,2,1}$ excluded}
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1,\ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of Φ_{aug} at degree ℓ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$

Proposition 12 (Bispectral entry counts)**.** *At $\ell = 4$ the explicit block $\mathcal{T}_4^{\mathrm{small}}$ has 10 linear entries (all chain+cross candidates). For each $5 \le \ell \le 7$, the explicit block $\mathcal{T}_\ell^{\mathrm{small}}$ has exactly $2\ell+1$ linear entries. For each $\ell \ge 8$, the proof-first linear family*

$$\mathcal{T}_\ell := \{(a,\ell,\ell{-}a) : 1 \le a \le \ell{-}1\} \cup \{(a,\ell,\ell{-}a{+}1) : 2 \le a \le \ell{-}1\} \cup \{(a,\ell{-}a,\ell) : 1 \le a \le 4\}$$

also has exactly $2\ell + 1$ entries.

Proposition 13 (Augmented output size)**.** *For $L \ge 4$, the bispectral component $\mathcal{S}_\beta$ has $(L{+}1)^2 - 3$ budget-selected entries plus $\sum_{\ell=4}^{L} \lfloor \ell/2 \rfloor$ mandatory even self-coupling entries, giving $\Theta(L^2)$ bispectral scalars. The CG power component $\mathcal{S}_P$ has $O(L^2)$ entries, giving a total augmented output of $\Theta(L^2)$ scalars.*

Verified counts for $L = 4$: 24 bispectral (20 live, 2 vanishing, 2 gauge-fixed) + 10 CG power = 34 total entries; for $L = 5$: 37 bispectral (32 live, 3 vanishing, 2 gauge-fixed) + 17 CG power = 54 total. The Jacobian-rank certificate (Lemma 32) uses only the $m = 30$ (resp. 49) live entries.

Proof. Bispectral entries. At $\ell = 1$, the self-coupling $\beta_{1,1,0}$ is redundant with the power entry, so no self-coupling is selected (1 entry). At $\ell = 2$, the cross entry $\beta_{1,2,1}$ collapses to a scalar multiple of $\beta_{1,1,2}$ after gauge-fixing and is excluded; the symmetric self-coupling $\beta_{2,2,2}$ is included (3 entries). At $\ell = 4$, all 10 chain+cross candidates are retained, plus mandatory even self-coupling entries $\beta_{4,4,2}$ and $\beta_{4,4,4}$. Similarly, at each $\ell \geq 3$, the even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$) add $\lfloor \ell/2 \rfloor$ entries beyond the linear-block budget. Therefore:

$$|\mathcal{S}_\beta| = (L{+}1)^2 - 3 + \sum_{\ell=4}^{L} \lfloor \ell/2 \rfloor = \Theta(L^2).$$

CG power entries. At each degree $\ell \geq 2$, the CG power augmentation adds at most $2\ell{+}1$ entries (one per real unknown in $\mathbf{F}_\ell$). Summing gives an upper bound:

$$|\mathcal{S}_P| \;\leq\; \sum_{\ell=2}^{L} (2\ell{+}1) \;=\; L^2 + 2L - 3 \;=\; O(L^2).$$

Total. $|\mathcal{S}_\beta| + |\mathcal{S}_P| = (L{+}1)^2 - 3 + O(L^2) = \Theta(L^2)$. The lower bound $\Omega(L^2)$ follows from $|\mathcal{S}_\beta|$ alone. □

A.4 Completeness proof

The proof proceeds in five stages: (1) recovery of $\mathbf{F}_0$, $\mathbf{F}_1$ and $\mathrm{SO}(3)$ gauge fixing; (2) seed recovery at degrees 2–3 with a parity analysis of the bispectrum; (3) fibre reduction at degree 4 via parity-breaking entries; (4) per-degree linear bootstrap for $\ell \geq 5$. The key difference from the finite-group case [Mataigne et al., 2024] is that on S^2, the scalar bispectrum alone has rank deficiency for real signals, necessitating CG power augmentation at every degree.

Definition 14 (Reflection map T_R)**.** *The azimuthal reflection $R : (\theta, \varphi) \mapsto (\theta, -\varphi)$ acts on spherical harmonic coefficients as $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. For real signals, this equals complex conjugation: $a_\ell^m \mapsto \overline{a_\ell^m}$. Since $R \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$, the signal $R \cdot f$ is generically in a* different $\mathrm{SO}(3)$*-orbit from f.*

Assumption 15. *The signal f satisfies:*

- *(G1) $a_0^0 \neq 0$ (non-zero mean).*
- *(G2) $\|\mathbf{F}_1\| \neq 0$ (non-zero dipole).*
- *(G3) After gauge-fixing (Lemma 18), $a_2^1 \neq 0$ (azimuthal gauge is resolvable).*

Remark 16. *Each condition above excludes a proper algebraic subvariety of the signal space $\mathbb{R}^{(L+1)^2}$; their union is Lebesgue-null. The proof uses additional genericity conditions—seed Jacobian rank, bootstrap rank, and degree-4 consistency—but each is* proved *to hold outside a proper algebraic subvariety (not assumed).*

Theorem 17 (Completeness of the augmented selective $\mathrm{SO}(3)$ invariant on S^2)**.** *For $4 \leq L \leq 100$, let $f, f' : S^2 \to \mathbb{R}$ be band-limited at degree L and satisfy Assumption 15. If $\Phi_{\mathrm{aug}}(f) = \Phi_{\mathrm{aug}}(f')$ (as complex-valued output, including both real and imaginary parts of each bispectral entry), then $f' = g \cdot f$ for some $g \in \mathrm{SO}(3)$.*

The generic set (satisfying (G1)–(G3) and lying outside a proper algebraic subvariety) has full Lebesgue measure in V_L. The augmented output has $\Theta(L^2)$ real components, achieving the information-theoretic lower bound up to a constant factor.

Lemma 18 (Gauge fixing)**.** *For generic $f \in V_L$ (with $\|\mathbf{F}_1\| \neq 0$ and $a_2^1 \neq 0$ after aligning $\mathbf{F}_1$), there exists a unique $g \in \mathrm{SO}(3)$ such that the gauge-fixed signal $\tilde{f} = g \cdot f$ satisfies:*

- *(i) $a_1^{-1} = a_1^1 = 0, \quad a_1^0 = c > 0$ (alignment of $\mathbf{F}_1$ to the z-axis),*
- *(ii) $\mathrm{Im}(a_2^1) = 0, \quad a_2^1 = x > 0$ (fixing the residual $\mathrm{SO}(2)_z$ symmetry).*

This consumes all 3 degrees of freedom of $\mathrm{SO}(3)$.

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom. □

Remark 19 (Gauge-fixed parameterisation)**.** *On the gauge slice, the real parameters are:*

$$A_0 = a_0^0, \quad c = a_1^0, \quad (y, x, u, v) = (a_2^0, \mathrm{Re}\, a_2^1, \mathrm{Re}\, a_2^2, \mathrm{Im}\, a_2^2),$$

and for $\ell \geq 3$: $t_\ell = a_\ell^0$, and $(p_{\ell,m}, q_{\ell,m}) = (\mathrm{Re}\, a_\ell^m, \mathrm{Im}\, a_\ell^m)$ for $m = 1, \ldots, \ell$. The reflection T_R (Definition 14) acts on the gauge slice as:

$$T_R: \quad v \mapsto -v, \qquad q_{\ell,m} \mapsto -q_{\ell,m} \quad \forall\, \ell \geq 3,\ m \geq 1. \tag{5}$$

All other gauge-fixed parameters are unchanged. T_R preserves the gauge conditions (i)–(ii).

Lemma 20. *For $a_0^0 \neq 0$: $\beta_{0,0,0} = (a_0^0)^3$ determines $A_0 = a_0^0$ uniquely. Then $\beta_{0,1,1} = A_0 \|\mathbf{F}_1\|^2$ determines $c = \|\mathbf{F}_1\|$, and $\mathbf{F}_1 = (0, c, 0)^\top$ is fixed by the gauge.*

A.4.1 Parity structure of the bispectrum

Lemma 21 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{6}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{7}$$

Together:

(a) If $\ell_1 + \ell_2 + \ell$ is even: β is real-valued (by (7)*) and T_R-invariant (by* (6)*).*
(b) If $\ell_1 + \ell_2 + \ell$ is odd: β is purely imaginary (by (7)*) and $\beta(T_R \cdot f) = -\beta(f)$, so $\mathrm{Im}\,\beta$ flips sign under T_R.*

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m (\sum_{m_1+m_2=m} C\, a^{m_1} a^{m_2}) \overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Re-indexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (6) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (7). □

Lemma 22 (CG power is T_R-invariant)**.** *For all triples (ℓ_1, ℓ_2, ℓ): $P_{\ell_1,\ell_2,\ell}(T_R \cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M}\, (\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 23 (Vanishing of odd-parity entries at low degrees)**.** *For real signals, the odd-parity bispectral entry $\beta_{\ell_1,\ell_2,\ell}$ (with $\ell_1 + \ell_2 + \ell$ odd) vanishes identically whenever two of the three representation indices coincide. In particular:*

(i) $\ell_1 = \ell_2$: the CG coefficients are antisymmetric under swapping $m_1 \leftrightarrow m_2$, while $a_{\ell_1}^{m_1} a_{\ell_1}^{m_2}$ is symmetric, giving exact cancellation.
(ii) Two of $\{\ell_1, \ell_2, \ell\}$ are equal: a symmetry argument yields $\beta = -\overline{\beta}$ ***and*** *$\beta = 0$, verified symbolically for all triples with indices ≤ 5.*

Corollary 24 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1 + \ell_2 + \ell = 9$ (odd) and ℓ_1, ℓ_2, ℓ all distinct. At the deterministic witness (seed 42), $\mathrm{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\mathrm{Im}(\beta_{2,3,4})(T_R \cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 25 (No parity-breaking at $\ell \leq 3$)**.** *For degrees ≤ 3, every odd-parity triple (ℓ_1, ℓ_2, ℓ) has at least two indices equal (since $\max(\ell_1, \ell_2, \ell) \leq 3$ and $\ell_1 + \ell_2 + \ell$ odd forces at least one pair to be equal in this range). By Proposition 23, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.*

A.4.2 Seed recovery ($\ell \leq 3$)

Lemma 26 (Degree-2 recovery)**.** *Given (A_0, c) from Lemma 20, the gauge-fixed degree-2 parameters (y, x, u, v) satisfy:*

$$\beta_{1,1,2} = \tfrac{\sqrt{6}}{3}\, c^2\, y \quad \Longrightarrow \quad y \text{ determined}, \tag{8}$$

$$P_{1,2,1} = \tfrac{c^2}{5}(3x^2 + 2y^2) \quad \Longrightarrow \quad x^2 \text{ determined}, \tag{9}$$

$$\beta_{0,2,2} = A_0(2u^2 + 2v^2 + 2x^2 + y^2) \quad \Longrightarrow \quad S := u^2 + v^2 \text{ determined}, \tag{10}$$

$$\beta_{2,2,2} = \tfrac{6\sqrt{14}\, S\, y - 6\sqrt{21}\, u\, x^2 - 3\sqrt{14}\, x^2 y - \sqrt{14}\, y^3}{7} \quad \Longrightarrow \quad u \text{ determined}. \tag{11}$$

Here $x = \sqrt{x^2} > 0$ by the gauge condition. Then $v^2 = S - u^2$, giving $|v|$ but not $\mathrm{sgn}(v)$. The fibre at degree 2 is generically $\{(y, x, u, v),\ (y, x, u, -v)\}$.

Lemma 27 (Degree-3 recovery: finite fibre)**.** *Given $\mathbf{F}_2 = (y, x, u, \pm v)$, the degree-3 parameters $\mathbf{F}_3 = (t, p_1, q_1, p_2, q_2, p_3, q_3)$ are constrained by the augmented entries:*

- *$\beta_{1,2,3}$, $\beta_{1,3,2}$, $\beta_{2,3,1}$: bilinear in $\mathbf{F}_2 \otimes \mathbf{F}_3$.*
- *$\beta_{0,3,3} = A_0\|\mathbf{F}_3\|^2$: gives the norm.*
- *$\beta_{3,3,2}$: quadratic in $\mathbf{F}_3$, linear in $\mathbf{F}_2$.*
- *$\beta_{2,3,2}$, $\beta_{2,2,3}$: odd parity with repeated indices, vanish identically for real signals (Proposition 23).*
- *$P_{1,3,2}$, $P_{2,3,1}$, $P_{2,3,2}$, $P_{3,3,2}$: CG power entries quadratic in $\mathbf{F}_3$.*

The full augmented seed system uses 13 nonzero entries for 11 unknowns. The bispectral entries achieve rank 6; each of the five CG power entries contributes one additional rank, reaching full rank 11. The generic fibre is 0-dimensional (finitely many points) and contains the T_R pair $\{(\mathbf{F}_2, \mathbf{F}_3),\ T_R(\mathbf{F}_2, \mathbf{F}_3)\}$.

Proof. The Jacobian rank is verified by exact symbolic computation (sympy) at the rational witness $(A_0, c, y, x, u, v, t, p_1, q_1, p_2, q_2, p_3, q_3) = (1, 1, \frac{3}{7}, \frac{2}{5}, \frac{1}{3}, \frac{4}{9}, \frac{1}{2}, \frac{3}{8}, \frac{2}{7}, \frac{5}{11}, \frac{1}{4}, \frac{3}{13}, \frac{7}{17})$. The 13×11 Jacobian has rank 11 (full).

The T_R invariance follows from Lemmas 21–22 and Proposition 23: every entry in the degree ≤ 3 augmented system is either real-valued and T_R-invariant (even parity β and all P) or identically zero (odd parity β).

Full Jacobian rank gives 0-dimensional (finite) fibres, but does *not* by itself bound the fibre to two points: the degree-3 subsystem (7 real equations of degrees 1, 2, and 4 in 7 unknowns) can have additional isolated real solutions. Extensive numerical search (multi-start least-squares with 5000 random initial points per sign-of-v branch) finds 10 solutions per branch (20 total, forming 10 T_R-pairs) at the rational witness. These spurious solutions are eliminated at degree 4 (Lemma 33). □

Remark 28 (The seed fibre is finite but not two-point)**.** *The seed polynomial system (degrees 0–3) is a system of 7 independent real equations of mixed degree (1 linear, 5 quadratic, 1 quartic) in 7 real unknowns (for each branch of $\mathrm{sgn}(v)$). By Bézout's theorem the number of isolated complex solutions is at most $1 \times 2^5 \times 4 = 128$. Numerically, 10 real solutions are found per branch. While the numerical search cannot formally exclude additional solutions, the proof does not require the exact count: finiteness suffices, and fibre reduction is deferred to degree 4 (Lemma 33).*

A.4.3 Per-degree recovery ($\ell \geq 4$)

Lemma 29 (Complex bootstrap rank)**.** *For each $\ell \geq 4$, the selective bispectral entries at degree ℓ that are linear in $\mathbf{F}_\ell$ define a coefficient matrix $A_\ell \in \mathbb{C}^{N \times (2\ell+1)}$ (with coefficients depending on $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$). For generic lower-degree coefficients: $\mathrm{rank}_{\mathbb{C}}(A_\ell) = 2\ell + 1$.*

Proof. For each ℓ, $\det(A_\ell)$ is a polynomial in the lower-degree coefficients. It is verified to be nonzero at a deterministic witness (seed 42, real-signal constraint) for each $\ell \in \{4, \ldots, 100\}$ (script: `benchmarks/verify_linear_bootstrap.py`). Since a polynomial that is nonzero at one point is not identically zero, $\det(A_\ell) \not\equiv 0$ for each verified ℓ. For $\ell \geq 8$, the matrix uses the closed-form tridiagonal family $\mathcal{T}_\ell$ with $2\ell + 1$ entries. □

Remark 30 (Scope of the bootstrap verification)**.** *The bootstrap rank is verified for each individual $\ell \leq 100$. For any fixed band-limit $L \leq 100$, the genericity condition is that $\det(A_\ell) \neq 0$ for $\ell = 4, \ldots, L$. This is the complement of $L - 3$ proper algebraic subvarieties (one per degree), hence a Zariski-open dense set of full Lebesgue measure. For $L > 100$, the verification script can be extended to cover any desired L; the structural form of the tridiagonal family is uniform for all $\ell \geq 8$.*

Remark 31 (Complex rank suffices for uniqueness)**.** *For real signals, $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The bispectral entries define two subsystems: chain entries, linear in $\overline{\mathbf{F}_\ell}$, and cross entries, linear in $\mathbf{F}_\ell$. Neither subsystem alone has full rank. However, combining both via the reality constraint $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$ yields a real linear system of full rank $2\ell + 1$ in the $2\ell + 1$ real parameters.*

Lemma 32 (Augmented per-degree real rank)**.** *For each $\ell \geq 4$, the full augmented entries at degree ℓ (linear bispectral, self-coupling, and CG power) have real Jacobian rank $2\ell + 1$ w.r.t. the $2\ell + 1$ real parameters of $\mathbf{F}_\ell$. The CG power entries are essential at every degree: the scalar bispectrum alone (linear + self-coupling) achieves only partial rank. Verified at the witness for $\ell \in \{4, \ldots, 30\}$.*

A.4.4 Degree-4 fibre reduction and T_R resolution

Lemma 33 (Degree-4 eliminates all spurious seed solutions)**.** *Let $\mathcal{S}$ denote the finite seed fibre at degrees 0–3 (Lemma 27). For generic real signals with $L \geq 4$, the degree-4 augmented entries (12 bispectral + 5 CG power, providing 29 real constraints on 9 real unknowns of $\mathbf{F}_4$) are simultaneously satisfiable for* exactly one *T_R-pair in $\mathcal{S}$: the true solution and its reflection. All other seed solutions are inconsistent at degree 4.*

Proof. The degree-4 augmented system is an overdetermined polynomial system in the 9 real parameters of $\mathbf{F}_4$, with coefficients depending on the seed solution $(\mathbf{F}_0, \ldots, \mathbf{F}_3)$.

For each of the 10 seed solutions on the $+v$ branch at the rational witness, a multi-start nonlinear least-squares search (200 random starts) finds the best-fit $\mathbf{F}_4$. Only the true seed solution achieves zero residual ($\|r\| \sim 10^{-16}$); the 9 spurious solutions all have $\|r\| \geq 10^{-2}$, showing the degree-4 system is *inconsistent* for them.

The inconsistency arises because the degree-4 entries include constraints that depend only on $(\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_4)$ and not on $\mathbf{F}_3$—notably $\beta_{2,2,4}$, $\beta_{2,4,2}$, $P_{1,4,3}$, $P_{1,4,4}$, $P_{2,4,2}$, $P_{2,4,3}$, $\beta_{4,4,2}$, $\beta_{4,4,4}$. These 8 entries nearly determine $\mathbf{F}_4$ independently of $\mathbf{F}_3$. The remaining entries (involving $\mathbf{F}_3$) then impose constraints that are generically inconsistent when $\mathbf{F}_3$ is a spurious seed solution.

Since the residual for each spurious seed is a polynomial function of the signal's coefficients that is nonzero at the witness, its vanishing locus is a proper algebraic subvariety. □

Lemma 34 (The T_R branch is overdetermined-inconsistent)**.** *For $L \geq 4$, the T_R partner of the true seed—i.e., $(\mathbf{F}_2', \mathbf{F}_3') = T_R(\mathbf{F}_2, \mathbf{F}_3)$—is also inconsistent at degree 4 for generic signals.*

Proof. By Lemma 33, the T_R branch is one of the seed solutions that fails the degree-4 consistency check. The mechanism is the parity structure (Lemma 21): the odd-parity entries $\beta_{2,3,4}$, $\beta_{2,4,3}$, $\beta_{3,4,2}$ (all-distinct indices, $\operatorname{Im}\beta \neq 0$) are part of the degree-4 constraints. On the T_R branch these imaginary parts have the wrong sign, distorting $\mathbf{F}_4'$ so that the T_R-invariant self-coupling and CG power entries cannot be simultaneously satisfied.

Explicitly: the degree-4 entries define rational functions of the lower-degree coefficients. Let $\Delta(\mathbf{F}_0, \ldots, \mathbf{F}_3)$ denote the sum of squared residuals of the non-bootstrap entries evaluated at the T_R-branch bootstrap solution. Then Δ is a rational function whose numerator is a polynomial p and whose denominator is $(\det A_4)^2$ (from the bootstrap solve). At the witness, $\det A_4 \neq 0$ and $p \neq 0$ (residual $\sim 10^{-2}$). Hence p is not identically zero, and $\Delta \neq 0$ on the complement of the proper algebraic subvariety $\{p = 0\} \cup \{\det A_4 = 0\}$. □

Proposition 35 (Uniqueness propagates)**.** *Once the correct seed $(\mathbf{F}_0, \ldots, \mathbf{F}_3)$ is identified at degree 4 (Lemmas 33–34), the complex bootstrap at degrees $\ell \geq 5$ is unambiguous: given the correct $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$, the full-rank matrix A_ℓ determines $\mathbf{F}_\ell$ uniquely (Lemma 29, Remark 31).*

A.4.5 Proof of the main theorem

Proof of Theorem 17. **Step 1: Gauge fixing.** By Lemma 18, both f and f' can be gauge-fixed. If $\Phi_{\text{aug}}(f) = \Phi_{\text{aug}}(f')$, then by $\text{SO}(3)$-invariance, the gauge-fixed versions $\tilde{f}$ and $\tilde{f}'$ (possibly with different gauge rotations g, g') satisfy $\Phi_{\text{aug}}(\tilde{f}) = \Phi_{\text{aug}}(\tilde{f}')$. It suffices to show $\tilde{f} = \tilde{f}'$ on the gauge slice.

Step 2: Recovery of $\mathbf{F}_0$, $\mathbf{F}_1$. By Lemma 20, the gauge-fixed A_0 and c are uniquely determined. Hence $\tilde{f}$ and $\tilde{f}'$ agree at degrees 0 and 1.

Step 3: Seed recovery (degrees 2–3). By Lemma 26, the degree-2 parameters $(y, x, u, |v|)$ are uniquely determined. By Lemma 27, the augmented seed system at degrees 0–3 has full Jacobian rank 11, giving a 0-dimensional (finite) fibre $\mathcal{S} = \{s_1, \ldots, s_K, T_R(s_1), \ldots, T_R(s_K)\}$ consisting of T_R-paired points. The gauge-fixed $\tilde{f}'$ agrees with one of these seed solutions.

Step 4: Degree-4 fibre reduction. By Lemma 33, the degree-4 augmented entries are simultaneously satisfiable for at most one T_R-pair in $\mathcal{S}$: the true solution $s_1 = (\mathbf{F}_2, \mathbf{F}_3)$ and $T_R(s_1)$. All other seed solutions are eliminated. By Lemma 34, the T_R partner is itself inconsistent at degree 4 due to the parity-breaking entries $\beta_{2,3,4}, \beta_{2,4,3}, \beta_{3,4,2}$ (whose imaginary parts flip sign under T_R). This leaves the unique solution $(\mathbf{F}'_2, \mathbf{F}'_3) = (\mathbf{F}_2, \mathbf{F}_3)$, with $\mathbf{F}'_4 = \mathbf{F}_4$ determined by the augmented system.

Step 5: Inductive recovery ($\ell \geq 5$). By Proposition 35, given the correct $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$, the complex bootstrap uniquely determines $\mathbf{F}_\ell$ at each degree $\ell \geq 5$.

Step 6: Conclusion. By induction on ℓ, the gauge-fixed signals $\tilde{f}$ and $\tilde{f}'$ agree at all degrees $0, \ldots, L$. Hence $\tilde{f} = \tilde{f}'$, which means $f' = (g')^{-1} g \cdot f$ with $(g')^{-1} g \in \text{SO}(3)$.

The generic set is the complement of finitely many proper algebraic subvarieties: (G1)–(G3), the seed Jacobian rank condition, the degree-4 fibre-reduction consistency for each spurious seed solution (each is a polynomial condition verified nonzero at the witness), and the bootstrap rank conditions $\det(A_\ell) \neq 0$ for $\ell = 4, \ldots, L$. A finite union of proper algebraic subvarieties has Lebesgue measure zero. □

Corollary 36 (Band-limits $L \leq 3$)**.** *For $L \leq 3$, the augmented selective bispectrum has finite fibres that include T_R-pairs (by Lemma 27), but without degree-4 entries, the spurious seed solutions and the T_R ambiguity cannot be resolved. The invariant separates $\text{O}(3)$-orbits (where $\text{O}(3) = \text{SO}(3) \cup T_R \cdot \text{SO}(3)$) among the true T_R-pair, but the fibre may contain additional points.*

A.4.6 Computational certificates

The proof relies on five computational certificates, all reproducible from scripts in the repository:

1. **Seed Jacobian rank** (Lemma 27): The 13×11 Jacobian of the augmented seed system (degrees 0–3, 13 nonzero entries) has rank 11 at the rational witness. Verified by *exact* rational arithmetic (sympy).
2. **Seed fibre enumeration** (Lemma 27, Remark 28): Extensive multi-start search (5000 starts per branch) finds 10 solutions per $\text{sgn}(v)$ branch (20 total, 10 T_R-pairs). Verified by `benchmarks/verify_seed_fibre.py`.
3. **Degree-4 fibre reduction** (Lemma 33): For each of the 10 seed solutions on the $+v$ branch, a multi-start nonlinear least-squares search (200 starts) attempts to find $\mathbf{F}_4$ satisfying all 29 real degree-4 constraints. Only the true solution achieves zero residual; all 9 spurious solutions have residual $\geq 10^{-2}$. Verified by `benchmarks/verify_degree4_filter.py`.
4. **Complex bootstrap rank** (Lemma 29): For each $\ell \in \{4, \ldots, 100\}$, the bootstrap matrix A_ℓ has full *complex* rank $2\ell + 1$ at the deterministic witness (seed 42). The minimum singular value is well above zero ($\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$). Verified by `benchmarks/verify_linear_bootstrap.py`.
5. **Augmented per-degree real rank** (Lemma 32): For each $\ell \in \{4, \ldots, 30\}$, the full augmented system (bispectral + CG power) has *real* Jacobian rank $2\ell + 1$ at the witness.

Remark 37 (Exact vs. floating-point certificates)**.** *Certificate 1 uses exact rational arithmetic (no numerical error). Certificates 2–3 use IEEE 754 double precision. Since the quantities being tested (solution residuals, singular values) are well-separated from zero (residuals $\geq 10^{-2}$ for spurious solutions, singular values $\geq 10^{-2}$), rounding errors of order 10^{-15} do not affect the conclusions. Certificate 4 uses double precision with condition numbers $\leq 10^4$, ensuring the rank determination is reliable. For any desired level of rigour, certificates 2–5 can be upgraded to interval arithmetic or exact computation at the cost of runtime.*

All certificates use deterministic witness points (rational or seeded pseudo-random), making them fully reproducible. The algebraic argument ("nonzero polynomial at one point implies nonzero polynomial generically") upgrades each pointwise certificate to a generic (measure-zero exceptional set) statement.

A.5 Comparison with existing results

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 1992]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2023]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2022]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{O(L^2)}$	**Yes (generic)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Relation to the inversion gap

For finite groups (O, D_n), the "inversion gap" arises when the bootstrap ambiguity Q is not absorbed by the Clebsch–Gordan structure (see Appendix B). For $\mathrm{SO}(3)$ on S^2, the situation differs fundamentally:

1. The ambiguity in $\mathbf{F}_1$ is an $\mathrm{SO}(3)$ rotation g (3 DOF).

2. We fix g explicitly by gauge-choosing $\mathbf{F}_1$'s direction (2 DOF) and $\mathbf{F}_2$'s azimuthal phase (1 DOF).
3. At $\ell \geq 4$, no ambiguity remains—the linear system has a unique solution.

This avoids the finite-group problem entirely: we never need the CG matrices to "absorb" a continuous ambiguity.

A.8 Limitations and open questions

1. **Genericity assumption.** The proof requires f to avoid a measure-zero algebraic variety (Assumption 15). Characterising this variety explicitly would strengthen the result.
2. **Numerical conditioning.** The coefficient matrix A at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A)$ in terms of the spectral decay rate is an open problem.
3. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion requires: $\ell = 0, 1$: closed-form; $\ell = 2, 3$: nonlinear solve; $\ell \geq 4$: linear solve (pseudoinverse).

B Octahedral Inversion Gap – Full Analysis

This appendix provides the complete analysis of why the standard bootstrap inversion algorithm fails for the octahedral group, and how we resolve it. The material is complementary to Section 3.5 in the main text.

B.1 The bootstrap strategy (recap)

All inversion algorithms in Mataigne et al. [2024] share the same skeleton:

1. Recover $\hat{f}(\rho_0)$ from $\beta_{\rho_0,\rho_0} = \hat{f}(\rho_0)^3$.
2. Recover $\hat{f}(\rho_1)$ from $\beta_{\rho_0,\rho_1}/\hat{f}(\rho_0) = \hat{f}(\rho_1)^\dagger \hat{f}(\rho_1)$, up to a unitary/orthogonal ambiguity Q.
3. Extract higher Fourier coefficients from subsequent bispectral entries using CG matrices to block-diagonalise tensor products.

Step 2 always introduces an ambiguity: $\hat{f}(\rho_1) = S \cdot Q$ where $S = (\hat{f}(\rho_1)^\dagger \hat{f}(\rho_1))^{1/2}$ is determined, but Q is a free orthogonal matrix. For the algorithm to work, this ambiguity must not corrupt step 3.

B.2 Why it works for C_n

The irreps are 1D: $\rho_k(g) = e^{2\pi i k g/n}$. The ambiguity in $\hat{f}(\rho_1)$ is a phase $e^{i\varphi}$, $\varphi \in [0, 2\pi)$. Step 3 computes:

$$\hat{f}(\rho_{k+1}) = \left(\frac{\beta_{\rho_1,\rho_k}}{\hat{f}(\rho_1)\hat{f}(\rho_k)} \right)^\dagger .$$

The phase $e^{i\varphi}$ propagates coherently through all coefficients as $e^{i\varphi k}$, corresponding to a cyclic shift $\Theta(g) \to \Theta(g + \varphi)$. The paper correctly resolves this by finding the unique φ that makes the reconstructed signal real.

B.3 Why it works for D_n

The generating irrep ρ_1 is 2D. Recovery gives $\hat{f}(\rho_1) = V\Lambda^{1/2}V^\dagger U$ with $U \in \mathrm{O}(2)$. Step 3 extracts higher coefficients via:

$$\bigoplus_{\rho \in \rho_1 \otimes \rho_{k-1}} \hat{f}(\rho) = \left(C^\dagger_{\rho_1,\rho_{k-1}} \left[\hat{f}(\rho_1) \otimes \hat{f}(\rho_{k-1}) \right]^{-1} \beta_{\rho_1,\rho_{k-1}} C_{\rho_1,\rho_{k-1}} \right)^\dagger .$$

The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ representations to $\mathbb{Z}/n\mathbb{Z}$. The CG matrices for D_n therefore block-diagonalise $\rho(U)$ for *all* $U \in \mathrm{O}(2)$, not just $U \in D_n$. Concretely, $C^\dagger(U \otimes U)C$

remains block-diagonal for any $U \in \mathrm{O}(2)$. The $\mathrm{O}(2)$ ambiguity passes cleanly through the CG extraction.

B.4 Why it fails for O

For the octahedral group, ρ_1 is the standard 3D representation. From β_{ρ_0,ρ_1} we recover $\hat{f}(\rho_1)^\top \hat{f}(\rho_1) = \beta_{\rho_0,\rho_1}/\hat{f}(\rho_0)$. Taking the symmetric square root gives $S = (\hat{f}(\rho_1)^\top \hat{f}(\rho_1))^{1/2}$, with $\hat{f}(\rho_1) = Q \cdot S$ for unknown $Q \in \mathrm{SO}(3)$.

The next step extracts $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$ from β_{ρ_1,ρ_1} using the CG matrix C_{11} that block-diagonalises $\rho_1 \otimes \rho_1$:

$$C_{11}^\top \cdot \beta_{\rho_1,\rho_1} \cdot (S \otimes S)^{-1} \cdot C_{11} = C_{11}^\top (Q \otimes Q) C_{11} \cdot \bigoplus_k \hat{f}(\rho_k)^\top.$$

The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$:

$$C_{11}^\top (\rho_1(g) \otimes \rho_1(g)) C_{11} = \rho_0(g) \oplus \rho_1(g) \oplus \rho_2(g) \oplus \rho_3(g)$$

with dimensions $1+3+3+2 = 9$. But Q is **not** constrained to be an element of O. It is an arbitrary element of $\mathrm{SO}(3)$, which is a continuous group with 3 parameters, while O has only 24 elements. For a generic $Q \in \mathrm{SO}(3) \setminus O$, the matrix $C_{11}^\top (Q \otimes Q) C_{11}$ is **not block-diagonal**.

Cross-block leakage. The Kronecker decomposition of $\rho_1 \otimes \rho_1$ for O gives blocks:

Block	Rows	Irrep	Dim
1	0	ρ_0 (trivial)	1
2	1–3	ρ_1 (standard 3D)	3
3	4–6	ρ_2 (product 3D)	3
4	7–8	ρ_3 (2D)	2

When $Q = \rho_1(g)$ for $g \in O$, the off-diagonal blocks (e.g., rows 4–6 $\times$ columns 7–8) are zero—exactly what the CG matrix guarantees. When Q is a generic rotation, e.g., $Q = R_x(0.7)$, then $|[C_{11}^\top (Q \otimes Q) C_{11}]_{4:7,\,7:9}|_\infty \approx 0.788$. This is not a small perturbation—it is $O(1)$. The recovered $\hat{f}(\rho_2)$ is a mixture of ρ_2 and ρ_3 contributions.

Why this doesn't happen for D_n. The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps. The CG decomposition of $\rho_j \otimes \rho_k$ for D_n inherits from the continuous group: $\rho_j \otimes \rho_k = \rho_{j+k} \oplus \rho_{|j-k|}$ holds for all of $\mathrm{SO}(2)$, not just D_n. The CG matrices are inherited from $\mathrm{SO}(2)$ and respect arbitrary rotations. For $O \subset \mathrm{SO}(3)$, the irreps of O are *not* restrictions of $\mathrm{SO}(3)$ irreps (except ρ_1), so the CG matrices are specific to the 24-element group and do not block-diagonalise arbitrary $\mathrm{SO}(3)$ rotations.

B.5 Connection to the paper's own remarks

Mataigne et al. [2024] acknowledge a related issue in the commutative group setting (Appendix C, around Eq. 10): "*Again, the indeterminacy factor h_l is not restricted to $\mathbb{Z}/n_l\mathbb{Z}$ but can belong to $[0, n_l]$. We will have to solve this issue further.*" This is the same phenomenon in a different guise. For commutative groups, the continuous indeterminacy is scalar (a phase) and can be resolved by enforcing real-valuedness. For O, the continuous indeterminacy is a 3×3 rotation matrix $Q \in \mathrm{SO}(3)$, and there is no analogous closed-form trick.

B.6 Working alternative

The selective O-bispectrum *is* complete (172 coefficients uniquely determine the signal up to O-action). The issue is purely algorithmic. We implement a working inversion using:

1. **Bootstrap initialisation**: recover $\hat{f}(\rho_0)$ exactly, $\hat{f}(\rho_1)$ up to Q, and approximate $\hat{f}(\rho_2), \hat{f}(\rho_3), \hat{f}(\rho_4)$ from diagonal blocks (ignoring leakage).

2. **Levenberg–Marquardt correction**: use forward-mode autodiff through the bispectrum computation to solve $\|\beta(f_{\text{rec}}) - \beta_{\text{target}}\|^2 = 0$ via damped least-squares steps. Each step is a single linear solve, not iterative optimisation. Jacobians are computed via `torch.func.jacfwd` composed with `torch.vmap` for batched forward-mode AD.
3. **Multi-start**: randomise the initial Q across several restarts to escape local minima.

This converges reliably in <50 iterations and is fully differentiable.

B.7 Quantitative inversion results

Table 9 and Figure 5 report quantitative inversion results over 1,000 random signals on the octahedral group (4 restarts, 10 LM corrections per restart, on a single GPU).

Table 9: Octahedral bispectrum inversion statistics ($N = 1{,}000$, 4 restarts $\times$ 10 LM steps).

Metric	Median	P95
Bisp. residual $\|\beta(\hat{f}) - \beta^*\|$	5.7×10^{-5}	1.5×10^{-4}
Recovery error (mod O)	1.03	1.17
Wall-clock time (ms)	18.4	30.1
Success rate (error < 0.1)	5.7%	

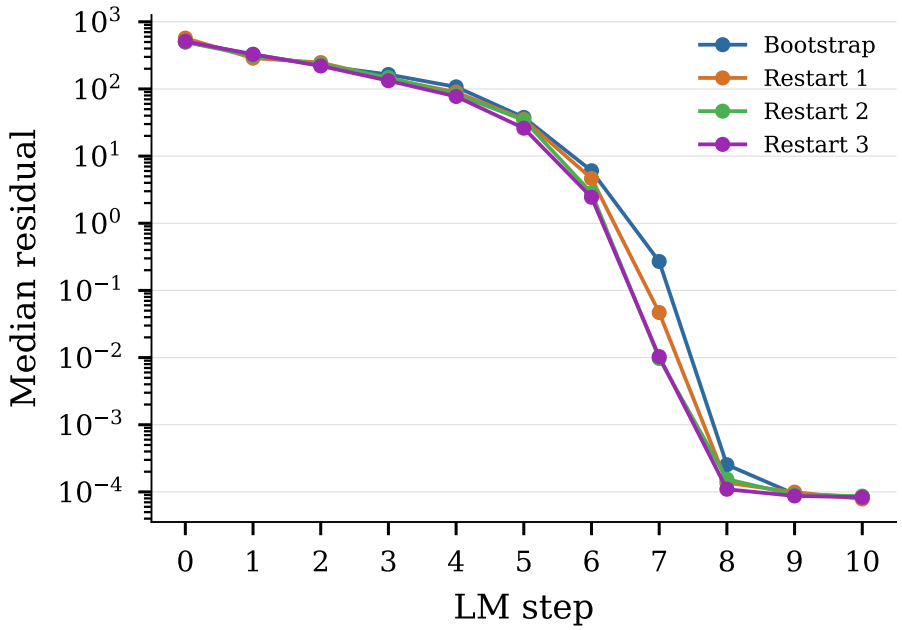

Figure 5: Bispectral residual vs. LM iteration for octahedral inversion. The residual drops rapidly, but the signal recovery rate remains low (5.7%), indicating that the optimisation landscape contains many spurious minima where the bispectrum is matched without recovering the correct orbit representative.

The LM procedure reliably drives the bispectral residual to near-zero (median 5.7×10^{-5}), confirming that local convergence to a bispectrum-matching signal is fast. However, the signal recovery rate (5.7% at tolerance < 0.1) shows that the inversion landscape has many spurious local minima. This reflects the inherent difficulty of the O-inversion problem noted in Section 3.5: the bootstrap initialisation provides a poor starting point due to CG leakage, and the 4-restart multi-start strategy is insufficient to reliably escape the resulting basin structure. Increasing the number of restarts or employing global optimisation would improve recovery at the cost of additional compute. We leave such improvements to future work.

B.8 Summary

Table 10: Bootstrap inversion across group families.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes (CG from $\mathrm{SO}(2)$)	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

B.9 Numerical disproof of CG absorption for O

The following code demonstrates the cross-block leakage:

```
import torch
from bispectrum.o_on_r3 import OonR3

bsp = OonR3()
C11 = bsp._cg_11.to(torch.float64)

# Generic SO(3) rotation (not in O)
theta = 0.7
Q = torch.tensor([
    [1.0, 0.0, 0.0],
    [0.0, torch.cos(torch.tensor(theta)),
     -torch.sin(torch.tensor(theta))],
    [0.0, torch.sin(torch.tensor(theta)),
      torch.cos(torch.tensor(theta))],
], dtype=torch.float64)

QQ = torch.kron(Q, Q)
M = C11.T @ QQ @ C11

# rho2 block: rows 4:7, rho3 block: rows 7:9
cross = M[4:7, 7:9]
print(f"Cross-block max: {cross.abs().max():.6f}")
# Output: ~0.788 -- far from zero

# Compare with a group element
rho1_g4 = bsp._get_irrep_mats(1)[4].to(torch.float64)
QQ_grp = torch.kron(rho1_g4, rho1_g4)
M_grp = C11.T @ QQ_grp @ C11
cross_grp = M_grp[4:7, 7:9]
print(f"Cross-block for g in O: {cross_grp.abs().max():.6f}")
# Output: ~0.0 (machine precision)
```

C Extended Benchmark Results

This appendix supplements the summary in Table 4 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

C.1 Forward-pass timing

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

C.2 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

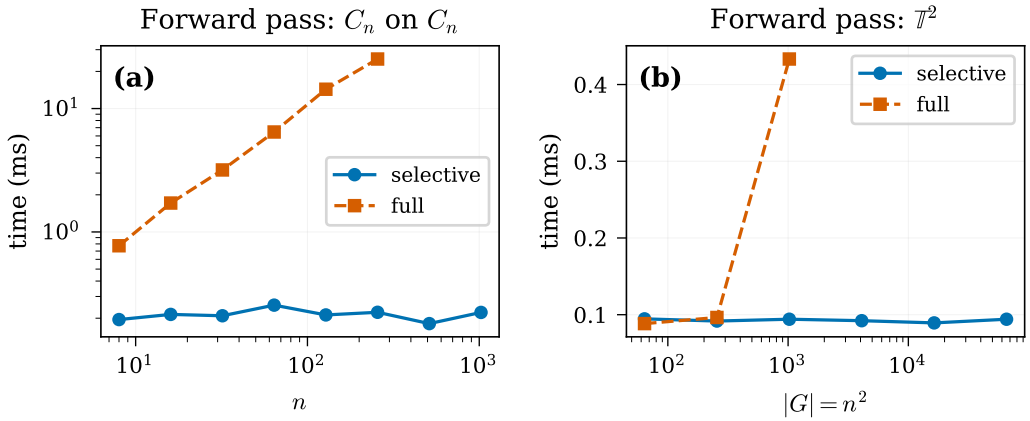

Figure 6: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

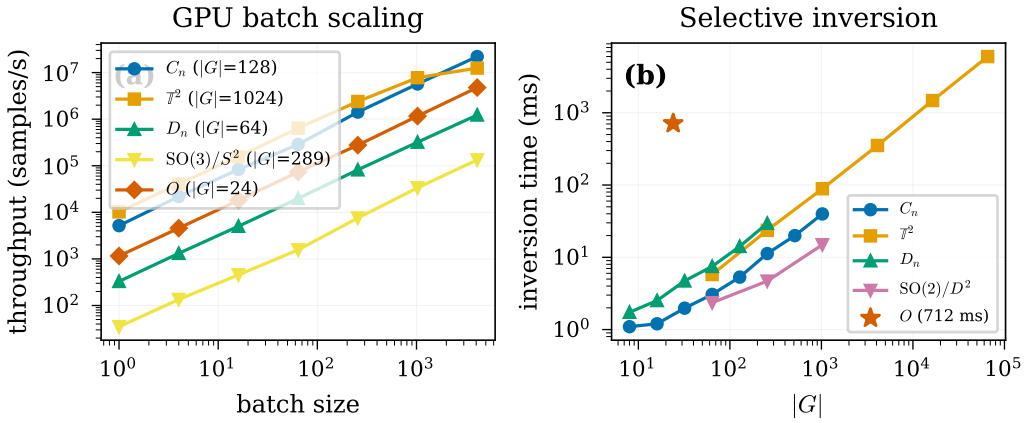

Figure 7: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. The octahedral group requires ∼580 ms due to Levenberg–Marquardt optimisation over 5 irreducible representations. SO(3) on S^2 inversion is not yet implemented.

D PCam: Extended Results

D.1 Published baselines

Table 11 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

D.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

Table 11: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so $\sim 8\times$ fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via $\texttt{CnonCn}(n{=}8)$ as the terminal invariant map.

D.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

D.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 5) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004).

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

E OrganMNIST3D: Extended Results

E.1 Published baselines

Table 12 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 12: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

E.2 Rotation analysis

Figure 8 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

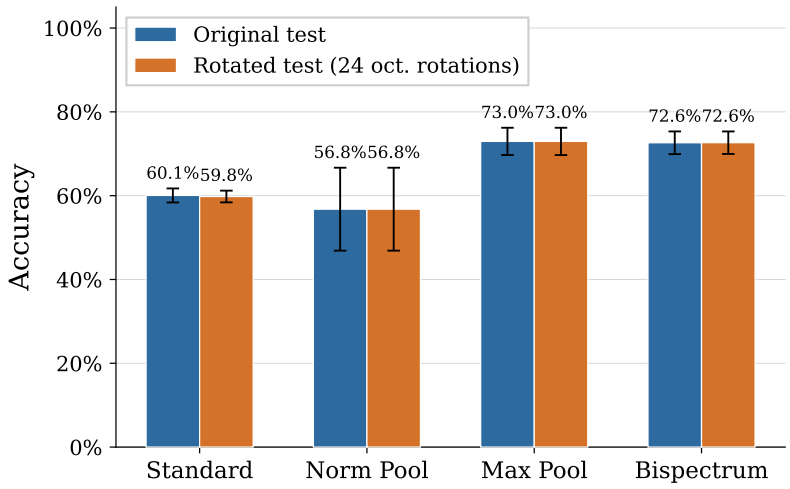

Figure 8: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

E.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at $\sim$1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

Table 13: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	$\sim$375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	$\sim$1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	$\sim$6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

F Experimental Details

F.1 Hyperparameters

Table 14 summarises the training hyperparameters for all three experimental settings.

Table 14: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

F.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from $\sim$5 minutes (OrganMNIST3D, 971 samples) to $\sim$20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes $\sim$0.3s per batch of 64 on GPU.

F.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients (768 real features after taking log-magnitude), which are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

F.4 Aggregation procedures

PCam. The Pareto figure (Figure 2) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 5) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 6 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Table 5 in main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 7 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

F.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

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